DATA LINEAGE REPORT

Enterprise Human Capital Data Management Platform

UC17 Enterprise Data Engineering Capstone

Document detail	Value
Organisation / case study	ABC Global Enterprises Ltd.
Document purpose	Technical record of source-to-target data movement and warehouse lineage
Covered ingestion scope	13 documented source datasets and Pentaho transformations
Warehouse model reference	Star-schema diagram supplied with the project documentation
Technology context	Pentaho Data Integration (Spoon), PostgreSQL, Python/Pandas, Power BI, Git/GitHub
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1. Lineage Purpose and Boundary

Data lineage records where data originates, which integration steps handle it, where it is stored, and which downstream warehouse objects and reports use it. In EHCMDP, lineage is maintained at three connected levels: dataset/table lineage, column-level mapping, and run-level operational lineage.

1.1 What this report covers

- Thirteen source datasets documented for Sprint 1, including their source format, Pentaho transformation name, processing steps and PostgreSQL staging target.
- The logical path from each staging target into the warehouse star schema and onward to Power BI.
- Required evidence for keys, relationships, data-quality decisions, row-count reconciliation and refresh status.

1.2 Boundary and evidence rule

The source-to-staging mappings below come from the project's Dataset Ingestion Documentation. The star-schema diagram is treated as the warehouse model reference. A diagram or transformation name alone does not prove that a job ran successfully or that every source is already loaded into a particular fact or dimension. Actual implementation must be confirmed using Pentaho logs, PostgreSQL catalog/row counts, SQL mappings and the Power BI model.

2. End-to-End Lineage Path

Stage	Lineage responsibility	Evidence to retain
1. Source dataset	Identify file/system, format, delivery or extract, and source record count.	Source inventory; file name/path; extract date; source row count.
2. Pentaho ingestion	Read the source using the appropriate input step and apply configured selection/filter steps.	KTR name/version; input-step settings; filter conditions; job log.
3. PostgreSQL staging	Write records to the documented staging table, retaining source-level identifiers where available.	Schema/table name; load timestamp; loaded row count; error output.
4. Validation and standardization	Check required fields, types, keys, duplicates and domain values before warehouse use.	Validation results; rejected/excluded counts; correction record.
5. Star-schema load	Resolve business keys to dimension keys and load the correct fact/dimension grain.	SQL/Pentaho mapping; primary/foreign keys; load strategy; reconciliation.
6. Power BI semantic model	Use curated warehouse tables and defined relationships/measures for report visuals.	Model relationship settings; DAX/measure definitions; refresh history.

2.1 Simplified lineage sequence

SOURCE FILES → PENTAHO KTRs → PostgreSQL staging.* → VALIDATION / KEY LOOKUPS → WAREHOUSE FACTS & DIMENSIONS → POWER BI

Python/Pandas profiling can support validation and reconciliation; Git/GitHub stores versioned KTRs, SQL, mappings and documentation. These are supporting controls, not separate business data destinations.

3. Sprint 1 Source-to-Staging Register

The register below reflects the supplied ingestion documentation. Source row counts are documented source volumes; they are not asserted to be the actual loaded PostgreSQL counts.

#	Source dataset	Format	Source rows	Cols	Pentaho transformation	Documented step sequence	PostgreSQL staging target
1	Employee Performance & Productivity	CSV	100,000	20	tr_employee_performance_ingestion.ktr	CSV File Input → Select Values → Table Output	staging.stg_employee_performance

2	Human Resource Dataset	CSV	8,950	15	Human_Resource_CSV_Ingestion.ktr	CSV File Input → Select Values → Filter Rows → Table Output	staging.stg_human_resource
3	HR Employee	CSV	1,470	23	HR_Employee_CSV_Ingestion.ktr	CSV File Input → Select Values → Filter Rows → Table Output	staging.stg_hr_employee
4	AI Fair Recruitment Dataset	JSON	121,200	16	AI_Fair_Recruitment_JSON_Ingestion.ktr	JSON Input → Select Values → Filter Rows → Table Output	staging.stg_ai_fair_recruitment
5	Payroll Data	Excel	3,216	47	Payroll_Data_Ingestion.ktr	Microsoft Excel Input → Select Values → Table Output	staging.stg_payroll
6	Employee Attendance Ratings	XML	477	101	Employee_Attendance_XML_Ingestion.ktr	XML Input Stream (StAX) → Filter Rows → Select Values → Table Output	staging.stg_attendance_xml
7	Employee Leave Tracking	CSV	300	11	Employee_Leave_CSV_Ingestion.ktr	CSV File Input → Select Values → Table Output	staging.stg_leave
8	HR Performance Rating	CSV	6,709	11	HR_Performance_Rating_CSV_Ingestion.ktr	CSV File Input → Select Values → Filter Rows → Table Output	staging.stg_performance_rating
9	Amazon Employee Access	JSON	32,769	11	Amazon_Employee_Access_JSON_Ingestion.ktr	JSON Input → Select Values → Table Output	staging.stg_access_control
10	HR SaaS Employee Master	Excel	793	38	HR_SaaS_Employee_Master_Excel_Ingestion.ktr	Microsoft Excel Input → Select Values → Table Output	staging.stg_hr_saas_employee_master
11	HR SaaS Employee Snapshot	XML	18,360	44	HR_SaaS_Employee_Snapshot_XML_Ingestion.ktr	XML Input Stream (StAX) → Filter Rows → Table Output	staging.stg_hr_saas_snapshots

12	HR SaaS Leave Requests	JSON	902	5	HR_SaaS_Leave_Request_JSON_Ingestion.ktr	JSON Input → Select Values → Table Output	staging.stg_hr_saas_leave_requests
13	Employee Performance Behavior	JSON	5,000	10	Employee_Performance_Behavior_JSON_Ingestion.ktr	JSON Input → Select Values → Table Output	staging.stg_performance_eval_behavior

3.1 Source-format summary

Source format	Dataset count
CSV	5
JSON	4
Excel	2
XML	2
Total	13

The sum of the retained documented source row counts is 264,153. Use this only as a source-volume reference; filtering, rejected records, duplicate handling and reruns may make staging counts different.

3.2 Source-specific lineage cautions

- Employee Attendance Ratings: the 101-column structure is Date plus Person_0 through Person_99. Confirm whether the warehouse expects this wide structure or a normalized employee/date attendance grain.
- Employee identifiers differ in name and format across datasets (for example Employee_ID, employee_id, EmployeeID and Employee ID). Do not join these fields directly without documented normalization and an approved identity crosswalk.
- Filter Rows appears in some transformations. Record the exact predicate and count the accepted and excluded records; do not treat source row count as guaranteed staging count.

4. Star-Schema Lineage

The star-schema diagram supplied with the project documentation is included here as the warehouse relationship reference. It shows several workforce fact tables connected to shared descriptive dimensions. Facts represent process-specific records or measurements; dimensions provide reusable context for analysis.

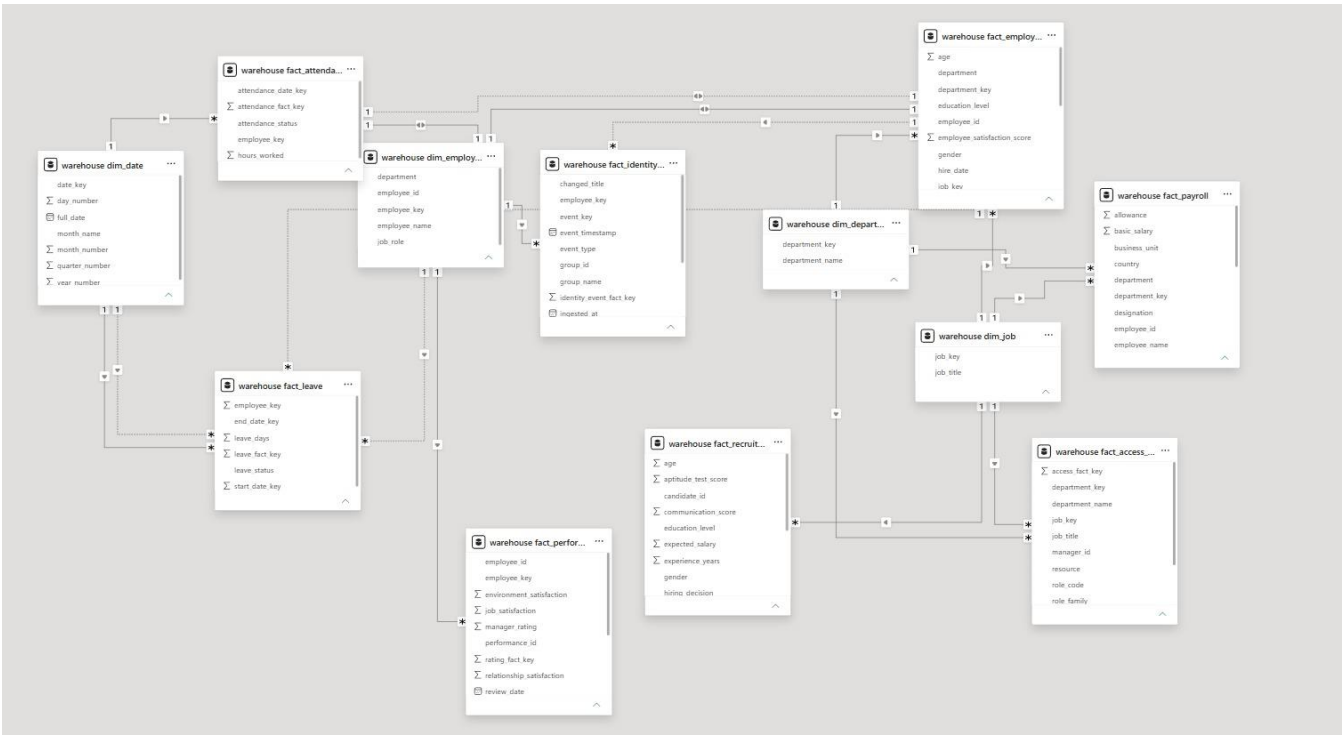


Figure 1. Supplied PostgreSQL / Power BI warehouse star-schema relationship diagram (rendered in grayscale).

Diagram object group	Purpose in reporting	Lineage details to verify
Date dimension	Provides calendar attributes for period filtering and time-based comparisons.	Date key; actual date field used; role of hire/review/start/end/payroll date.
Employee dimension	Provides consistent employee-level descriptive context.	Enterprise employee key; source IDs; duplicate/collision handling; history strategy.
Department dimension	Supports department-level slicing and roll-ups.	Department key; canonical department values; handling of unknown/unmatched departments.
Job dimension	Supports role/job-level grouping.	Job key; title normalization; effective dates if job titles change.
Attendance fact	Stores attendance-related events or measures at its implemented grain.	Grain; employee/date foreign keys; duplicate event rule; hours/duration definitions.
Leave fact	Stores leave requests, days or related leave measures.	Request/event grain; employee and date roles; leave type and approval status.

Payroll fact	Stores payroll values and compensation components.	Employee/period grain; currency; earnings/deductions; reconciliation rules.
Recruitment fact	Stores recruitment applications/events or related measures.	Candidate vs application grain; requisition/stage keys; cohort and date definitions.
Performance fact	Stores performance reviews, ratings or measures.	Review-cycle grain; employee/date keys; rating-scale compatibility.
Identity/access fact or table	Represents identity/access records shown in the diagram.	Exact physical table, restricted fields, identity mapping and permitted analytics use.

The diagram contains physical table labels and fields, but lineage should not infer hidden relationships from visual proximity alone. Confirm the actual PostgreSQL keys and Power BI relationship cardinalities before declaring a relationship implemented.

4.1 How staging data maps to the warehouse

Staging source group	Potential warehouse subject	Mapping rule before loading
Employee Performance & Productivity; Human Resource; HR Employee; HR SaaS Employee Master; HR SaaS Snapshot	Employee dimension and relevant workforce/performance facts, as supported by implemented model.	Resolve source employee IDs through the approved crosswalk; define whether each row is current-state or a dated snapshot.
Payroll Data	Payroll fact and applicable employee/organization/date dimensions.	Define payroll period grain, monetary component inclusion, currency and employee key lookup.
Employee Attendance Ratings	Attendance fact if its source semantics support the attendance measures.	Confirm the meaning of Person_0…Person_99, date grain and whether data must be unpivoted.
Employee Leave Tracking; HR SaaS Leave Requests	Leave fact and employee/date dimensions.	Map employee identity, leave type, approval status and start/end dates; define day-count rules.
AI Fair Recruitment Dataset	Recruitment/candidate-related fact or target defined by the model.	Preserve candidate/application identifiers and distinguish hiring decision from application stage.
HR Performance Rating and Performance Behavior	Performance fact(s) or separate subject tables, depending on the implemented grain.	Do not merge different score types without aligned employee keys, review period and rating definitions; use compatible employee keys, review periods and rating definitions.
Amazon Employee Access	Identity/access subject table or fact shown in the model.	Keep access fields restricted; use only approved identity crosswalks and documented target mapping.

5. Source-to-Warehouse Mapping Rules

Mapping element	Required lineage detail
Source field	Exact file/sheet/path and original column name.
Staging field	Exact PostgreSQL staging table and column after Select Values / input parsing.
Transformation	Pentaho KTR and relevant step; include field rename, cast, filter, calculation or lookup.
Business key	Source identifier and its scope; state if it is unique only within that source.
Warehouse key	Target dimension surrogate key or fact foreign key; identify lookup logic.
Target grain	What one row represents in the fact/dimension, including event, employee, review or period.
Quality rule	Required/non-null checks, valid ranges, duplicate rule, reference checks and rejected-record handling.
Load behavior	Insert, update, upsert or snapshot strategy; duplicate prevention and rerun behavior.
Downstream consumer	Power BI table/relationship, measure or visual using the target object.

5.1 Rules that prevent incorrect joins

- Normalize whitespace and formatting of employee IDs, but do not remove leading zeros or alter case unless the source contract permits it.

- Use a source-to-enterprise employee crosswalk when identifiers are not guaranteed to be globally unique.
- Document join type, join columns, expected cardinality, effective-date rules and treatment of unmatched records.
- Keep different date meanings separate—for example, hire date, payroll period, attendance date, leave start/end and performance review date.
- Do not join fact tables directly to one another simply because both contain an employee ID; use shared dimensions and preserve each fact's grain.

6. Operational Lineage Evidence

Run metadata	What to capture
Run identity	Unique run ID, Pentaho job/KTR name and version or Git commit.
Source evidence	Source filename/path or source object, format, modified/extract timestamp and input count.
Execution evidence	Start/end timestamps, run status, step-level errors and retry information.
Staging outcome	Target table, inserted/updated row counts, rejected count and final table count.
Warehouse outcome	Fact/dimension target, load mode, key-lookup failures, inserted/updated counts and reconciliation status.
Validation outcome	Null/duplicate/domain/reference checks, exceptions, owner and resolution status.
BI outcome	Power BI dataset/report name, refresh timestamp/status and validation of key measures.

Do not populate run timestamps, successful statuses or row counts from estimates. Record actual values from Spoon execution results, PostgreSQL queries and Power BI refresh history.

7. Data Quality and Reconciliation Points

Boundary	Minimum checks	Evidence
Source → Pentaho	File exists/readable; expected structure; expected columns; parse errors.	Input log, schema check and source row count.
Pentaho → staging	Selected fields/types valid; filter criteria documented; no silent column shifts.	KTR step log, error rows, staging row count.
Staging → dimensions	Required business keys; duplicate policy; canonical categories; unknown member handling.	Profiling output, key lookup results, duplicate report.
Staging → facts	Fact grain; foreign keys; valid dates; measure ranges; no accidental row multiplication.	SQL reconciliation and referential-integrity checks.
Payroll reporting	Totals by approved period and currency; gross/net calculation consistency.	Source-to-target count and amount reconciliation.
Attendance / leave	Date ranges, durations, leave status and employee link consistency.	Exception list and aggregate reconciliation.
Power BI model	Relationship direction/cardinality, refresh completion and measure totals.	Model validation and refresh evidence.

7.1 Reconciliation formula

Source rows = accepted rows + rejected rows + documented exclusions

Where a transformation updates or merges records, record inserted and updated counts separately. Reconcile staging counts to warehouse loads using the documented load strategy rather than assuming one source row always becomes one fact row.

8. Change Impact and Traceability

Change introduced	Lineage questions to check	Likely downstream scope
Source column renamed/removed	Which KTR input step and Select Values step refer to it? Which staging columns and mappings use it?	Affected staging table, warehouse load and BI fields/measures.

Employee ID format changed	Does the crosswalk still resolve the key? Are leading zeros or source prefixes significant?	Employee dimension and all facts joined through employee keys.
Filter predicate changed	How many rows will be newly included/excluded? Does the business definition change?	Staging volume, fact counts and any dashboard totals.
Dimension attribute changed	Is history retained? Do existing fact keys still point to the correct version?	Every fact and visual that uses the shared dimension.
Fact grain or relationship changed	Could joins multiply rows or change aggregation meaning?	Measures, totals, drill-down behavior and report visuals.
Power BI measure changed	Is its source field and calculation still aligned with the approved definition?	Affected reports and interpretation of KPIs.

For each change, update the mapping register, transformation metadata, test evidence and documentation in the same Git change. Use a commit or release reference to connect the document to the implemented code.

9. Implementation Verification Checklist

- ☐ All 13 retained source datasets and corresponding KTR files are present in the repository.
- ☐ Each KTR has been executed and its success/failure evidence retained.
- ☐ Each of the 13 retained staging targets exists in PostgreSQL schema staging.
- ☐ Actual source and staging row counts have been recorded for the same run.
- ☐ Filter Rows predicates and excluded/rejected counts are documented for applicable KTRs.
- ☐ Employee identifiers and cross-source key resolution rules are approved and tested.
- ☐ Employee Attendance Ratings wide structure has been reviewed for intended warehouse grain.
- ☐ Fact and dimension names, grains, primary keys and foreign keys match the live PostgreSQL model.
- ☐ Power BI relationships and cardinalities match the star-schema diagram and live model.
- ☐ Core counts and measures are reconciled between source, staging, warehouse and Power BI.
- ☐ Final document and mapping metadata are committed with the release.

10. Lineage Outcome

This report provides a technical trace from the 13 retained Sprint 1 source datasets into their PostgreSQL staging targets and defines the evidence needed to trace those records through the warehouse star schema to Power BI. Its main purpose is traceability: a reviewer should be able to identify the source, transformation, target, key rule, quality outcome and downstream consumer for each important field or metric.

The report should be considered implementation-complete only when the actual Pentaho execution logs, PostgreSQL object/row-count evidence, warehouse source-to-target mappings and Power BI relationship checks are attached to or referenced by the project repository. No appendix section is included in this report.